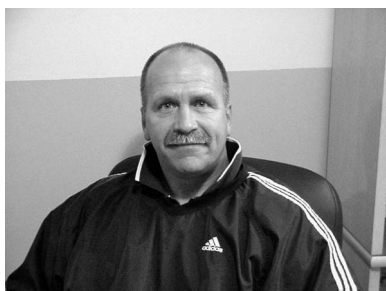


The Assisted Jump Squat: An Alternative Method for Developing Power in Adolescent Athletes

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SUMMARY

THE ASSISTED JUMP SQUAT CAN PROVIDE A SAFE AND EFFECTIVE MEANS OF INCREASING LOWER LIMB POWER IN CONJUNCTION WITH TRADITIONAL TRAINING METHODS. BAND ASSISTANCE MAY ALSO SHIFT THE MOVEMENT SPECIFICITY PROFILE OF THE JUMP SQUAT EXERCISE CLOSER TO ACTUAL SPORTING MOVEMENTS BY INCREASING MOVEMENT VELOCITIES ATTAINED.

POWER TRAINING: TRADITIONAL METHODS

Power has traditionally been developed in the weight room setting using weightlifting and plyometrics (10). Although these are proven methods for developing lower

limb power production, they do have a number of limitations (10). Weightlifting can be difficult to learn and require expert coaching to achieve high levels of proficiency. Until the athlete has mastered the lifts, the power production benefits of these exercises are limited because of the light loads that must be handled during initial drilling and until the athlete has learned how to explosively recruit the muscles of the hip, knee, and ankle in a coordinated fashion. Although the weightlifting lifts are among the most explosive movements in sport, the velocity of movement is relatively slow. Garhammer (5) has reported barbell velocities of 1.6–2.0 m/s for maximum effort cleans and snatches in elite lifters. If we consider that elite high jumpers can achieve center of mass vertical velocities of up to 4.8 m/s at takeoff (8), it would seem obvious that other training methods may also be required to target the velocity-specific needs of certain sports.

Plyometrics are an effective training method used to bridge the gap between strength and speed (1) and have been found to increase power outputs and improve jump performance and running velocity in athletic populations (12). However, although plyometrics are an excellent method for developing power for athletes, they require very good force production and force reduction capabilities. Without a solid

strength base, medium- or high-intensity plyometric exercises may be unsafe for adolescent athletes (4). The large ground reaction forces inherent in plyometrics mean that unless they are used judiciously within a program, they can lead to acute overload injuries and chronic overuse problems, particularly in growing athletes (4).

POWER TRAINING: COMMON ALTERNATIVE METHODS

Another effective special exercise for developing explosive power in the lower limbs that is sometimes underused is the jump squat (Figure 2). Jump squats do not require the same high level of coordination as weightlifting or plyometrics. Once the back squat has been mastered and a base level of strength is developed, most athletes have no problem performing this movement effectively. A common error with execution in weightlifting is failure to fully extend and finish the pull, but this is not an issue with the jump squat. Because it is a jumping movement, complete triple extension is ensured. With the jump squat, coaches must be very mindful of the bar loads used. Although elite athletes in contact sports have been reported to routinely use loads greater than 60% of 1 repetition maximum on this movement (2), programming similar relative loads in high school athletes may greatly increase the risk of injury. Heavy loads on the jump squat can easily lead to

a breakdown in technique, which may have negative consequences for the knees or spine of an athlete. Furthermore, research has found that power output is inversely related to barbell load in this exercise—the lighter the load, the greater the power production because of increased movement velocity (3). Therefore, it would appear unnecessary to have athletes perform heavily weighted jump squats if power production is the primary reason for selecting this exercise (3). This is supported by previous research that has demonstrated that training at light loads on jumps squats leads to adaptations at low-load high-velocity movements, whereas training at heavy loads impacts force development qualities (6).

Variable resistance training is another method sometimes used to stimulate strength and power adaptations (11). Using this method, athletes will typically add resistance to the movement being trained using bands. Bands are useful because they alter resistance patterns through the range of motion, ensuring that athletes must produce as much effort at mechanically strong positions as they do in mechanically weak ones. This phenomenon has been termed “accommodating resistance” (13). However, by switching the band’s

direction of pull upwards (Figure 1), we have found a new novel way of combining bands and the jump squat exercise to produce an interesting new training stimulus, which may have particular relevance for young athletes.

PRACTICAL APPLICATIONS—THE ASSISTED JUMP SQUAT

Instead of following the traditional approach of resisting the movement performed, we have found that using bands to assist in the performance of a jump squat can be a very effective method of developing lower limb power in elite junior track and field athletes. During the squat descent, the band serves to unload the bar, so the athlete has less force to overcome at the weakest point along the jumping range of motion. This unloading also takes a lot of the stress off the knees and back for athletes, making it a safer exercise for young athletes to perform. However, one of the most interesting aspects of this exercise is the altered velocity profile that results from using bands to assist the movement. The greater the assistance used, the greater the movement velocity and resultant jump height. In essence, this method forces the athlete to produce force at faster velocities than they normally

would in the weight room setting. It is an adaptation of the commonly used assisted sprint training method. The Table highlights the movement velocity alterations that can be achieved by manipulating barbell assistance and how it compares with some traditional training methods and loads. Programming specific band types to manipulate jump velocity can assist the strength and conditioning coach to try and match the specificity of the training stimulus to the demands of the sport. High-velocity exercises may have an appropriate place in a periodized strength training program designed for athletes who require speed and quickness (9). Furthermore, a physiological window of trainability for developing speed strength has been identified to occur between the ages of 14 and 18 years for adolescent athletes (7). The assisted jump squat may provide a useful method to take advantage of this sensitive training window.

Although this method may have a role to play in the preparation of elite senior athletes also, to date, we have found it to be a very effective and novel training method for high school athletes. Training with stronger bands results in very fast movement speeds and exaggerated jump heights, which young athletes find both challenging and fun (Figure 3). With enough band assistance, athletes can actually perform jumps with a resistance below that of their body weight! By providing specific feedback, such as jump height or movement velocity, through means of a linear position transducer, a great sense of competition can be added, which young athletes enjoy, ensuring maximal efforts during training sets. A total of 3–5 sets of 1–5 repetitions are recommended.

SAFETY

When performing assisted jump squats, safety is a very important concern for the strength and conditioning coach. Assisted jumps can be performed in either a Smith machine or a power cage. The advantage of the Smith machine is that it controls for any horizontal displacement ensuring

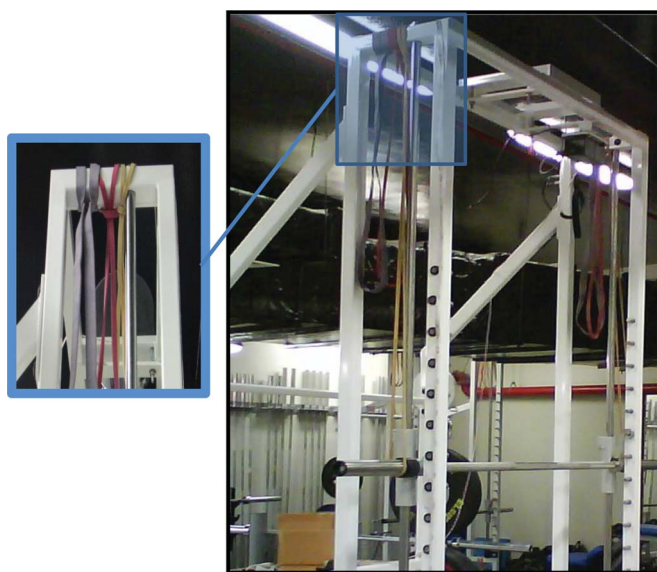


Figure 1. Assisted jump squat band setup.

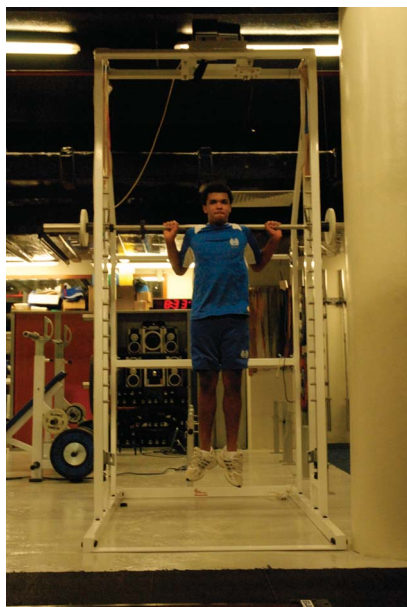


Figure 2. 30 kg resisted jump squat—maximum jump height.



Figure 3. Band-assisted jump squat—maximum jump height.

consistent landings and jump trajectories. However, if bands are aligned correctly and a spotter is employed behind the athlete to monitor and control horizontal displacement, it can be executed safely in a power cage. A lot of headroom is required above the athlete to jump into to ensure no injuries occur—we use and recommend 3-m-high cages to ensure adequate head clearance. If ample headroom is not

available in the Smith machine or power cage, then this exercise is contraindicated. Due care must also be taken when selecting band types. Too much band assistance can of course lead to excessive jump heights being achieved, which in turn can increase the chances of a potentially injurious landing. To date, we have used up to 3 mini bands at one time and have not had a single case of injury.

Table Example of velocity continuum of strength and jumping exercises		
Hypothetical velocity continuum		
Movement	Load (kg)	Peak velocity (m/s)
Squat	100	1.0
Weightlifting	100	1.6–2.0
Jump squat	40	2.2
Vertical jump (BW)	BW	3.0
Assisted jump squat (2 band)	BW—10 at bottom	3.2
Assisted jump squat (6 bands)	BW—30 at bottom	3.8
Elite high jump	BW off single leg	4.8
BW = body weight.		
Modified from Muller & Hommel and laboratory data.		

CONCLUSION

Assisted jump squats provide a safe and novel method of developing power in adolescent athletes. They should be used in conjunction with traditional training methods and may be of particular relevance during specific phases of preparation for sports requiring fast explosive lower limb movements.



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